

Insurance Data Analysis Project

Introduction to Insurance and Project Overview

What is Insurance?

Insurance is a financial product provided by insurance companies to safeguard individuals and entities against potential financial losses. These losses can arise from various unforeseen events such as accidents, natural disasters, theft, or health issues. In exchange for a premium paid by the policyholder, the insurance company commits to covering specific types of risks, thereby providing financial security and peace of mind.

Project Overview

This is a company specializing in offering a variety of insurance products to its customers. The company's portfolio includes auto insurance, which covers losses or damages related to vehicles. The provided dataset contains anonymized information about the policyholders, their vehicles, and their claim histories. This project analyzes this dataset to uncover insights that can help the company improve its services, optimize premiums, and reduce claim costs.

The company aims to enhance its auto insurance division by leveraging data analysis to:

1. Identify patterns in claim frequencies and amounts.
2. Understand the demographics and characteristics of policyholders.

Develop targeted marketing strategies to attract and retain customers.

Project Brief

You are tasked with analyzing the provided dataset to help this Insurance Company achieve the above objectives. Your analysis should be thorough and provide actionable insights. Use any data analysis tool of your choice (e.g., Python, Excel, SQL, Power BI) to conduct this analysis. The project should be completed within one week.

Key Questions to Answer

1. **Claim Frequency and Amount Analysis**
 - What are the average claim frequencies and amounts for different demographic groups (e.g., age, gender, marital status)?
 - Are there any specific vehicle characteristics (e.g., make, model, year) that correlate with higher claim frequencies or amounts?
2. **Risk Assessment**

- Which factors (e.g., household income, education level, coverage zone) are most indicative of high-risk policyholders?
- Can we identify any common characteristics among policyholders who make frequent claims?

3. Customer Segmentation and Marketing

- What are the key characteristics of policyholders with low claim frequencies and high household incomes?
- How can we segment the customer base to identify high-value customers for targeted marketing campaigns?

4. Demographic Analysis

- How does the distribution of policyholders vary across different demographic factors (age, gender, marital status)?
- Are there any noticeable trends in car usage and ownership among different demographic groups?

5. Geographical Analysis

- How do claim frequencies and amounts vary across different coverage zones?
- Are there any regional trends or patterns that should be taken into consideration for marketing or risk assessment?

6. Customer Behavior Insights

- How does the presence of children driving affect the frequency and number of claims?

Data Dictionary

Column Name	Description
ID	Unique identifier for the policyholder
birthdate	Birthdate of the policyholder
marital_status	Marital status of the policyholder (Single, Married, etc.)
car_use	Purpose of car use (Private, Commercial)
gender	Gender of the policyholder (Male, Female)
kids_driving	Number of kids of the policyholder who are driving
parent	Whether the policyholder is a parent (Yes, No)
education	Education level of the policyholder (High School, Bachelors, etc.)
car_make	Make of the policyholder's car
car_model	Model of the policyholder's car
car_color	Color of the policyholder's car
car_year	Year of the policyholder's car
claim_freq	Frequency of claims made by the policyholder
coverage_zone	Area of coverage (Highly Urban, Urban, Suburban, Rural, Highly Rural)
claim_amt	Amount claimed by the policyholder
household_income	Household income of the policyholder

Deliverables

1. Cleaned dataset.
2. Dashboard/Report
3. Detailed report answering the key questions.
4. Presentation slides summarizing key insights and recommendations.